AI PRODUCT MANAGEMENT CASE STUDY

Dispute-triage assistant for small merchants

Sebastian · October 2026 · Synthetic data and public card-scheme guidance only · Repository available on request

Problem. When a small merchant receives a card dispute (chargeback), they often don't have a clear understanding of the claim or of which evidence would win, so they accept the loss or submit evidence that doesn't answer the claim. This prototype reads the cardholder's complaint and returns the dispute category, the evidence to submit, and a short explanation that cites the rules, with a human-review flag when it isn't sure.

Decision: launch with **Claude plus human review** (98% accuracy, 0 confident errors, under 1p per dispute). **Agreement routing** between two local models matched Claude's accuracy at about 3% of the cost, but misses the evidence target, so it's the scale-up path, not the launch option. Full reasoning in Part 2.

Approach

1. **Define success first.** A PRD with targets set before any model ran: accuracy ≥ 85%, confident-but-wrong ≤ 5%, review rate ≤ 30%, evidence recall ≥ 80%.
2. **Build an eval set.** 80 synthetic disputes, labelled blind by me against a 6-category taxonomy built from public Visa and Mastercard guidance, with a written labelling rules log.
3. **Build the pipeline.** Retrieve rules (local embeddings) → classify with confidence → recommend evidence from a fixed list → explain with citations → apply the review rule in code.
4. **Compare and iterate.** Claude Sonnet 5.5 against Qwen 3.5 4B and 9B (local), all graded by the same scorer. Then two changes: prompt v2, and routing between models.

Results

	Claude	Qwen 4B (v2 prompt)	Qwen 9B (v1 prompt)	Agreement routing (4B + 9B, else Claude)	Target
Category accuracy	**98%**	96%	96%	**98%**	≥ 85%
Confident-but-wrong	**0%**	3.8%	3.8%	2.5%	≤ 5%
Evidence recall	**83%**	69%	70%	69%	≥ 80%
Latency median	**2.5s**	5.3s	11.7s	17.7s	
Cost per 1,000 cases	$8.57	£0	£0	**$0.29**	

What I learned

- **Accuracy didn't separate the models; safety did.** All three were 95–98% accurate in the first round, but only Claude's confidence tracked its errors. The local models gave every answer between 0.85 and 0.95 confidence, so a single review threshold caught almost nothing. **Thresholds must be calibrated for each model.**
- **Half of the first-round errors came from a gap in the spec, not the models:** complaints with two claims and no rule for which wins.
- **The same prompt fix helped the 4B and hurt the 9B** (96% → 89%). Kept as a negative result.
- **Two models agreeing is a better signal than one model's confidence.** But two models from the same family share blind spots.

Limitations

80 synthetic cases (one case = 1.25 points); written by Claude and labelled by one person; 3 labels corrected under written rules after the first runs (original-label results kept). Claude runs at its API default temperature; the Qwen models at 0. Local timings are from one laptop.

PART 2

Decision memo: dispute-triage assistant

Recommendation: ship **Option A, Claude with human review**, to a pilot group of small merchants. Plan **Option B, local agreement routing**, as the scale-up path once real merchant data confirms its signal.

All numbers: 80 synthetic cases, labelled by me, scored on corrected labels (results/comparison_v2.md, results/routing_v2.md). One case = 1.25 points of accuracy.

The decision

	A. Claude + review (launch)	B. 4B + 9B agree, else Claude (scale-up)
Category accuracy (ambiguous cases)	**98%** (92%)	**98%** (92%)
Confident-but-wrong (target ≤ 5%)	**0%**	2.5%
Sent to human review	14%	0%*
Evidence recall (target ≥ 80%)	**83%** ✓	69% ✗
Latency median / p90 (p90 target ≤ 10s Claude, ≤ 20s local)	**2.5s / 6.3s** ✓	17.7s / 23.4s ✗
Cost per 1,000 cases	$8.57	**$0.29**
Data leaving the device	100% of cases	**2%**

*B's review rule rarely fires, because the local models gave every answer between 0.85 and 0.95 confidence. Its safety comes from the agreement check, not from review.

Why A for launch: it's the only option that meets every PRD target, including evidence quality, which is what a merchant actually submits. It's also the fastest and the simplest to run. At **under 1p per dispute**, cost doesn't matter for a merchant with a few disputes a month.

Why not B yet: it matches A on category but **misses the evidence target** and the p90 latency target (23s against 20s), and its 98% rests on only 2 escalations (C044, C047). It also has **shared blind spots**: C048 and C056 got through because both local models were wrong in the same way. It becomes the better choice when volume, cost or data residency matter more, and once real data confirms the agreement signal.

Target merchants

Small UK online merchants (e-commerce, services and bookings, subscriptions) handling a few disputes a month without a payments or risk team. Not for high-volume merchants, who already use dedicated dispute tools, or mainly card-present businesses, where there's little evidence to recommend.

Human-review threshold

Send a case to a person if **any** of these is true: confidence < **0.70**, top two categories within 0.15, model says unclear, no rule section retrieved, or invalid output.

For Claude, 0.70 sits inside the safe range (0.50–0.87): **0 confident errors, 14% review**. Every Claude error in testing was flagged. **The threshold must be recalibrated for any other model.** The same rule caught almost none of the Qwen models' errors (0–5% review), because every Qwen answer scored 0.85 or above. The merchant always approves before anything is submitted. The tool recommends; it never files.

Failure modes seen in testing

1. **Complaints with two claims** ("poor service *and* the refund never came"): half of all first-round errors (8 of 16, on the original labels). Mitigated by a claim-priority rule (refund promised > recurring charge > the goods or service). The cost: the same rule made the 9B over-apply it (96% → 89%).
2. **Recurring vs one-off charges** (a second "pack" charged after notice).

3. **Evidence lists that are long and unfocused:** precision 38–50%. Merchants may submit irrelevant documents.
4. **Ambiguous labels:** 3 of 80 gold labels changed under written rules after the first runs. Real disputes will be messier.

Guardrails

- A fixed vocabulary of 26 evidence IDs (**0 invented IDs across all 400 answers in the five full runs**).
- Explanations must cite rule sections; citations to sections not provided are logged.
- JSON validation with one retry: **100% valid output**.
- Monitor weekly: review rate (alert above 30%), human overturn rate, and the share of multi-claim cases.

Data handling

Option A sends the complaint text, merchant type, amount and date to the Anthropic API; no card numbers or account details are needed. Before a pilot: strip customer names, emails and order numbers; confirm Anthropic's commercial data-retention terms; tell merchants in the terms of use. Option B keeps 98% of cases on the merchant's or provider's own hardware.

Provenance and licence

- **Models:** Claude Sonnet 5.5 (Anthropic, commercial API, $2 / $10 per million tokens, checked 5 Oct 2026); Qwen 3.5 4B and 9B (Alibaba Qwen team, Apache 2.0; 9B licence to confirm on its model card); nomic-embed-text v1.5 (Nomic AI, Apache 2.0).
- **Data:** synthetic cases only; rules summarised from public Visa and Mastercard guidance.

What would change my mind

- **Ambiguous-case accuracy below 80% on real data:** send all multi-claim complaints to a person by default.
- **Agreement routing at ≥ 97% accuracy on 300+ real cases, with both local models giving the same wrong answer in under 2% of cases, and evidence fixed:** switch to B.
- **A merchant or provider who can't send data to a US API provider:** B becomes the only option, with Claude replaced by human review for disagreements.

Next tests

1. **Real merchant disputes** (300+, anonymised), with a held-out test set; then a pilot measuring dispute win rate, merchant time per dispute and human overturn rate.
2. **Fix evidence:** Claude-generated checklists with local classification, or evidence-focused examples.
3. **Agreement routing across different model families**, to reduce shared blind spots.
4. **Claude with prompt v2**, to check whether the claim-priority rule helps or hurts the strongest model.

APPENDIX

PRD: Dispute-triage assistant for small merchants

Status: v1.1 (targets confirmed after the baseline run) · **Owned by:** Sebastian · **Date:** 5 Oct 2026 · Portfolio prototype using synthetic data and public card-scheme guidance only.

User

The owner or operations lead at a small online merchant: a few staff, no payments or risk team, and disputes handled between other jobs. They see a few chargebacks a month, through their payment provider's dashboard.

Problem

A dispute arrives as a reason code and the customer's short complaint, with a tight response deadline. The merchant doesn't know what they're actually being accused of or what evidence answers it. So they either accept the loss, or submit generic evidence that doesn't address the claim and lose anyway. Both outcomes cost revenue and add to the merchant's dispute ratio.

Job to be done

When a dispute lands, help me understand what I'm being accused of and exactly what evidence to send, so I can respond correctly before the deadline without becoming a card-scheme expert.

Scope

In: For each case, the system returns JSON with: the category (one of the 6 in docs/categories.md, or unclear), a confidence score, a checklist of recommended evidence, a plain-language explanation that cites the retrieved rule passages, and a human-review flag. Pipeline: retrieve rules (embeddings) → classify → recommend evidence → explain. Compares Qwen 3.5 4B, Qwen 3.5 9B (both local) and Claude.

Out: Submitting responses to providers; predicting win likelihood or advising whether to fight; fraud prevention; real merchant or cardholder data; live integration with payment providers; ATM, instalment and other out-of-scope dispute types; legal advice.

Success metrics

Measured per model on the labelled synthetic evaluation set: 80 cases (60 minimum), about 70% easy and 30% ambiguous, with at least 6 cases per category. Accuracy is calculated on cases the system does not mark unclear; those count as sent to review and are capped by the review-rate guardrail.

Metric	Definition	Target
Category accuracy	Predicted category = gold label; also reported separately for easy and ambiguous cases	**≥ 85%** overall
Evidence recall	Share of the gold evidence items for each case that appear in the recommendation, averaged across cases	≥ 80%
Latency	Median and p90 seconds from request to complete JSON	p90 ≤ 20 s local, ≤ 10 s Claude
Cost per case	API spend ÷ cases (also shown per 1,000 cases); local models = £0 marginal	≤ $0.02 Claude
Guardrail: confident-but-wrong rate	Share of cases *not* flagged for review that are wrong	≤ 5%
Guardrail: review rate	Share of cases flagged for human review	≤ 30%
Guardrail: valid JSON	Output passes the schema check (one retry allowed)	100%

Three targets were marked provisional in v1. After the baseline run **I kept them unchanged**, so no target was moved after seeing results.

Human-review rule

Flag `needs_human_review = true` if **any** of these is true:

1. The category is `unclear`.
2. Confidence is below **0.70**. This is a starting value; I'll tune it on the evaluation set until the confident-but-wrong rate is ≤ 5%.
3. The top two categories are within 0.15 of each other.
4. Retrieval returned no rule passage above the similarity threshold, so there's nothing to ground the answer on.
5. The output still fails the schema check after one retry.

Risks and mitigations

- **Wrong category leads to wrong evidence and a lost dispute.** Mitigation: the human-review rule, plus the confident-but-wrong guardrail.
- **Self-reported confidence from small models is poorly calibrated.** Mitigation: tune the threshold against observed accuracy rather than trusting the raw score; report a calibration table.
- **Invented rules or reason codes.** Mitigation: explanations must cite retrieved passages, and categories and evidence IDs are validated against fixed lists.
- **Synthetic data is cleaner than real complaints.** Mitigation: include the confusion pairs, short or vague narratives, and noise. Results don't claim to reflect real-world performance.
- **Scheme rules change.** Mitigation: sources are dated in `docs/categories.md`. This is a prototype, not operational guidance.
- **Local model limits on 24 GB.** Mitigation: one model loaded at a time, a fixed 8,192-token context, and thinking mode off.

Change log

- **v1.1:** provisional targets confirmed unchanged. Categories reduced from 7 to 6 (`authorisation_issue` merged into `processing_error` during labelling). Labelling rules added: received-or-used test, claim priority (H1), recurring signal (H2). See the labelling rules log in `docs/categories.md`.